

Paper B-07: Logistics and Supply-Chain Dynamics: A Constraint-Driven Flux Dynamics (CDFD) Perspective

Steve Bico Mujjabi, MD

Independent Researcher

Founder, Vura Labs

Kampala, Uganda

ORCID: 0009-0001-0556-5516

August 2026

Shared Part IV notation and evidence boundaries are defined in PART_IV_METHODS_AND_CLAIM_BOUNDARY.md. This paper states only its local observables, comparison, and failure conditions.

Abstract

This paper develops a falsifiable CDFD/AFL model of Logistics. It maps orders, shipments, materials, and customer demand to driving flux Φ , vehicle, warehouse, port, labor, and information capacity to active constraint C , rerouting, inventory, substitution, prioritization, and coordination to responsiveness S , and contracts, inventories, path dependence, and disruption history to structural memory M_s . The target outcome is lead time, fill rate, cost, shortage, and recovery, tested against shipment traces, inventory records, network maps, lead times, and disruption logs. The paper treats universality as a shared model architecture rather than an assertion that unlike systems have the same material mechanism. It states the negative result that would make the mapping unnecessary and places the domain claim inside the cumulative CDFD programme and its reproducible runtime boundary.

Consolidation Scope and Provenance

This consolidated paper incorporates the logistics and supply-chain scopes formerly carried by Papers B-07 and D-11. Inventory, transport, supplier disruption, and bullwhip effects are candidate measurable mechanisms, not consequences of shared notation.

1 Introduction

A supply chain is a physical transport network. Ore is mined in one hemisphere, processed in another, and consumed globally. To maximize profit, corporations have spent decades eliminating warehouses and stockpiles, relying on "just-in-time" (JIT) delivery.

In thermodynamic terms, JIT eliminates all buffer capacity from the system. It creates a network that is incredibly fast and efficient under perfect conditions but possesses zero topological elasticity. Under CDFD, this is a system balanced on a razor's edge, predicted here to be vulnerable to catastrophic failure upon the introduction of turbulence.

2 The Logistical Adaptive Surface

We evaluate supply chain networks using the Adaptive Operating Ratio:

$$\Psi_s = \left(\frac{\Phi}{C} \right) \cdot S \cdot M_s \quad (1)$$

For a specific logistical node (e.g., a major shipping port):

- **Flux** (Φ): The volume of cargo containers or tonnage of goods arriving and departing daily.
- **Constraint** (C): The physical unloading capacity of the cranes, labor availability, customs friction, and canal width restrictions.
- **Responsiveness** (S): The available warehouse buffer space and the ability to dynamically reroute ships to alternative ports.
- **Memory** (M_s): Fixed, long-term shipping contracts and consolidated mega-vessel dependencies. High memory locks the global shipping topology into a few rigid, highly efficient corridors, eliminating alternative pathways.

2.1 The Ischemia of Just-In-Time Logistics

A JIT supply chain intentionally drives responsiveness S (buffer stock) to zero to save money. To maintain $\Psi_s \approx 1$, the flow Φ and the constraint C must remain absolutely constant.

If a single major constraint spikes—such as a ship grounding in a critical canal or a sudden labor strike at a mega-port— $C \rightarrow \infty$ for that specific node. Because $S \approx 0$ and M_s is locked (ships cannot easily reroute or unload elsewhere), the adaptive ratio for the entire global segment plummets to $\Psi_s \ll 1$.

The flux Φ backs up. Ships idle in the ocean. Factories on the other side of the planet halt production because they lack buffer inventory. The localized blockage induces a global topological stroke. The system's failure to maintain structural elasticity results in a complete cessation of macroscopic thermodynamic throughput.

3 Measurement Layer

4 Cross-Domain Model Upgrade: Mechanism, Scale, and Tests

4.1 Observable Translation

The paper-specific translation begins with an observable chain rather than a verbal analogy. For Logistics, the proposed drive is orders, shipments, materials, and customer demand. The active limitation is vehicle, warehouse, port, labor, and information capacity. Responsiveness is carried by rerouting, inventory, substitution, prioritization, and coordination, while retained state is represented by contracts, inventories, path dependence, and disruption history. The outcome to explain is lead time, fill rate, cost, shortage, and recovery.

Table 1: Operational CDFD/AFL map for Paper B-07.

Term	Paper-specific observable meaning	Measurement question
Φ	orders, shipments, materials, and customer demand	What rate, load, gradient, or demand enters the focal system?
C	vehicle, warehouse, port, labor, and information capacity	Which measured bottleneck limits transfer, function, or recovery?
S	rerouting, inventory, substitution, prioritization, and coordination	Which process changes effective capacity or reroutes the load?
M_s	contracts, inventories, path dependence, and disruption history	Which retained state makes equal present loads produce different futures?
Y	lead time, fill rate, cost, shortage, and recovery	Which outcome is predicted out of sample, with uncertainty?

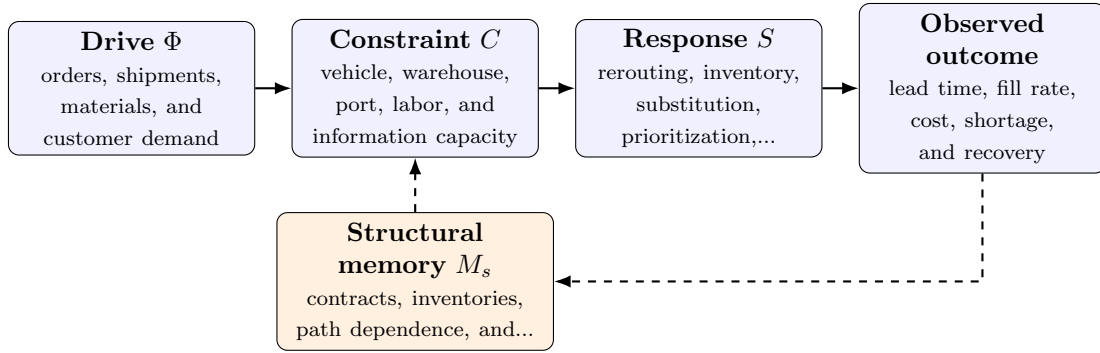


Figure 1: Paper B-07 observable CDFD/AFL architecture for Logistics. Solid arrows give the proposed forward chain; dashed arrows make history-dependent feedback explicit.

4.2 Paper-Specific Causal Hypothesis

For this paper, the hypothesized causal sequence is specific. A change in orders, shipments, materials, and customer demand arrives faster than rerouting, inventory, substitution, prioritization, and coordination can compensate. This raises or redistributes vehicle, warehouse, port, labor, and information capacity. If the load persists, the retained state encoded by contracts, inventories, path dependence, and disruption history changes the next response, so a later event of equal magnitude can produce a different trajectory in lead time, fill rate, cost, shortage, and

recovery. The scientific task is to estimate that sequence against the best ordinary model of the domain, not merely to redescribe the outcome after it occurs.

4.3 Cross-Scale Translation Without Mechanistic Erasure

For the engineered systems family, the cross-domain translation compares demand, designed capacity, feedback control, dependency topology, and accumulated technical state. The strongest engineering use is prospective: the mapping must identify a bottleneck, a control action, or a recovery signature before the failure occurs. [2, 4, 3, 1] The required scale declaration for this paper is hours to decades; parcels to global trade networks. At short scales, the key question is whether the incoming drive is transmitted, buffered, or diverted. At intermediate scales, response changes effective capacity and network exposure. At long scales, retained structure can alter the state space itself. A result is genuinely cross-scale only when the aggregation rule is stated and the same conclusion is not an artifact of averaging.

4.4 Evidence Ladder and Discriminating Tests

Table 2: Evidence ladder for Paper B-07. Each rung can fail independently.

Test	Design	Discriminating result
Proxy audit	Measure shipment traces, inventory records, network maps, lead times, and disruption logs and preregister how each record maps to Φ , C , S , M_s , and Y .	The mapping fails if proxies are non-identifiable, circular, or change meaning across cases.
Load ramp	Apply or observe a supplier or corridor loss during a demand surge while estimating the dominant constraint and response time.	A threshold claim requires a reproducible nonlinear change and uncertainty bounds, not a visually chosen breakpoint.
Recovery and memory	Match present load across units with different histories, then compare recovery paths.	The memory claim requires history-dependent divergence after current conditions are controlled.
Model comparison	Compare the baseline domain model with and without the CDFD/AFL state variables using held-out data.	The paper’s stated falsifier is: memory and topology terms do not improve shortage or recovery prediction.

4.5 Research Programme and Boundary Conditions

The immediate research programme has four steps. First, construct a documented dataset from shipment traces, inventory records, network maps, lead times, and disruption logs and retain raw units before normalization. Second, fit a baseline domain model and report its calibration and out-of-sample error. Third, add the smallest identifiable CDFD/AFL state extension and test whether it improves prediction of lead time, fill rate, cost, shortage, and recovery. Fourth, repeat the analysis across the declared scale range and publish negative cases as part of the cross-domain translation audit.

5 Conclusion

Supply chains are subject to a comparable flow problem to biological circulatory systems. A biological organism maintains fat reserves and redundant arteries (high S , low M_s) precisely to survive localized trauma. By analyzing global logistics through CDFD, we model the hypothesis that economic efficiency must be balanced with topological elasticity; otherwise, the network predicts its own eventual collapse.

Conflict of Interest

The author declares no conflict of interest.

Data Availability

Part-specific runtime diagnostics are under each Part's `outputs/` directory.

Release-wide code: `Part_E_Synthesis/supplementary/`.

Generated summaries: `Part_E_Synthesis/outputs/`.

Figures: `Part_E_Synthesis/figures/`.

6 Reference Layer

References

- [1] Per Bak, Chao Tang, and Kurt Wiesenfeld. Self-organized criticality: An explanation of the $1/f$ noise. *Physical Review Letters*, 59(4):381–384, 1987. DOI: <https://doi.org/10.1103/PhysRevLett.59.381>.
- [2] Albert-Laszlo Barabasi and Reka Albert. Emergence of scaling in random networks. *Science*, 286(5439):509–512, 1999. DOI: <https://doi.org/10.1126/science.286.5439.509>.
- [3] Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27(3):379–423, 1948. DOI: <https://doi.org/10.1002/j.1538-7305.1948.tb01338.x>.
- [4] Duncan J. Watts and Steven H. Strogatz. Collective dynamics of small-world networks. *Nature*, 393:440–442, 1998. DOI: <https://doi.org/10.1038/30918>.